Add Items in List

In [ ]:
```
1)append()
2)extend()
3)insert()
```

1.append()

In [ ]:
```
It is used to add items at last index.
Syntax:-List.append(value)
```

In [1]:
```
list1=[2,3,4,5,6]
list1.append(100)
list1
```

Out[1]:
```
[2, 3, 4, 5, 6, 100]
```

In [2]:
```
city_names=["Pune","Mumbai","Banglore","Delhi"]
city_names.append("Hydrabad")
city_names
```

Out[2]:
```
['Pune', 'Mumbai', 'Banglore', 'Delhi', 'Hydrabad']
```

In [3]:
```
city_names=["Pune","Mumbai","Banglore","Delhi"]
city_names.append({'a':97,'z':85})
city_names
```

Out[3]:
```
['Pune', 'Mumbai', 'Banglore', 'Delhi', {'a': 97, 'z': 85}]
```

In [4]:
```
list1=[2,3,4,5,6]
list1.append(200,300)
list1
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[4], line 2
      1 list1=[2,3,4,5,6]
----> 2 list1.append(200,300)
      3 list1

TypeError: list.append() takes exactly one argument (2 given)
```

In [6]:
```
list1=[2,3,4,5,6]
len(list1)
```

Out[6]:
```
5
```

2)extend()

In [ ]:
```
List.extend(iterable)
iterable>>List or tuple
```

In [7]:
```
list1=[2,5,4,6,7]
list2=[10,20,30,40]
list1.append(list2)
list1
```

Out[7]:
```
[2, 5, 4, 6, 7, [10, 20, 30, 40]]
```

In [9]:
```
list1=[2,5,4,6,7]
list2=[10,20,30,40]
list1.extend(list2)
print(list1)
print(list2)
```

```
[2, 5, 4, 6, 7, 10, 20, 30, 40]
[10, 20, 30, 40]
```

In [10]:
```
list1=[2,5,4,6,7]
list2=[10,20,30,40]
list2.extend(list1)
print(list1)
print(list2)
```

```
[2, 5, 4, 6, 7]
[10, 20, 30, 40, 2, 5, 4, 6, 7]
```

In [11]:
```
list1=[10,20,30,40]
list1.extend([1,2,3,4])
list1
```

Out[11]:
```
[10, 20, 30, 40, 1, 2, 3, 4]
```

In [12]:
```
list1=[1,2,3,4,5]
list2=[10,20,30,40,50]
list3=list1+list2
print(list1)
print(list2)
print(list3)
```

```
[1, 2, 3, 4, 5]
[10, 20, 30, 40, 50]
[1, 2, 3, 4, 5, 10, 20, 30, 40, 50]
```

In [13]:
```
x=[10,20,30,40,50]
y=(8,6,4,2)
x.extend(y)
print(x)
print(y)
```

```
[10, 20, 30, 40, 50, 8, 6, 4, 2]
(8, 6, 4, 2)
```

In [14]:
```
x=[10,20,30,40,50]
y=(8,6,4,2)
y.extend(x)
print(x)
print(y)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[14], line 3
      1 x=[10,20,30,40,50]
      2 y=(8,6,4,2)
----> 3 y.extend(x)
      4 print(x)
      5 print(y)

AttributeError: 'tuple' object has no attribute 'extend'
```

In [16]:
```
x=[10,20,30,40,50]
y=[8,7,6,5,4]
for i in y:
    x.append(i)
    print(x)
```

```
[10, 20, 30, 40, 50, 8]
[10, 20, 30, 40, 50, 8, 7]
[10, 20, 30, 40, 50, 8, 7, 6]
[10, 20, 30, 40, 50, 8, 7, 6, 5]
[10, 20, 30, 40, 50, 8, 7, 6, 5, 4]
```

3)insert()

```
Syntax: insert(index,value)
```

In [17]:
```
x=[2,3,4,5,6]
x.insert(1,500)
print(x)
```

```
[2, 500, 3, 4, 5, 6]
```

In [18]:
```
city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.append("Hydrabad")
city_names
```

Out[18]:
```
['Pune', 'Banglore', 'Mumbai', 'Delhi', 'Hydrabad']
```

In [19]:
```
city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.insert(2,"Hydrabad")
city_names
```

Out[19]:
```
['Pune', 'Banglore', 'Hydrabad', 'Mumbai', 'Delhi']
```

In [26]:
```
city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.insert(2,"Hydrabad")
city_names
```

Out[26]:
```
['Pune', 'Banglore', 'Hydrabad', 'Mumbai', 'Delhi']
```

Delete Items From List

In [ ]:
```
1)remove()
2)pop()
3)clear()
```

1)remove()

In [ ]:
```
Syntax:List.remove(Value)
```

In [27]:
```
city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.remove("Delhi")
city_names
```

Out[27]:
```
['Pune', 'Banglore', 'Mumbai']
```

In [28]:
```
city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.remove("DELHI")
city_names
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[28], line 2
      1 city_names=["Pune","Banglore","Mumbai","Delhi"]
----> 2 city_names.remove("DELHI")
      3 city_names

ValueError: list.remove(x): x not in list
```

In [29]:
```
city_names=["Pune","Banglore","Mumbai","Delhi"]
city="DELHI"
if city in city_names:
    city_names.remove(city)
    print("item deleted")
else:
    print("Item not found")
```

```
Item not found
```

2)pop():

In [ ]:
```
List.pop(index)
```

In [31]:
```
city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.pop(2)
city_names
```

Out[31]:
```
['Pune', 'Banglore', 'Delhi']
```

In [33]:
```
list1=[10,20,30,40,50]
list1.pop()
list1
```

Out[33]:
```
[10, 20, 30, 40]
```

In [34]:
```
list1=[10,20,30,40,50]
list1.pop(-1)
list1
```

Out[34]:
```
[10, 20, 30, 40]
```

In [35]:
```
list1=[10,20,30,40,50]
list1.pop(-2)
list1
```

Out[35]:
```
[10, 20, 30, 50]
```

3)clear()

In [ ]:
```
List.clear()
```

In [36]:
```
list1=[2,3,4,5]
list1.clear()
list1
```

Out[36]:
```
[]
```

del keyword

In [37]:
```
city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names[0]
```

Out[37]:
```
'Pune'
```

In [38]:
```
city_names=["Pune","Banglore","Mumbai","Delhi"]
del city_names[1]
city_names
```

Out[38]:
```
['Pune', 'Mumbai', 'Delhi']
```

7)index()

In [ ]:
```
index=List.index(value)
```

In [39]:
```
num=[20,30,40,50,60]
num.index(50)
```

Out[39]:
```
3
```

In [40]:
```
num=[20,30,40,50,70,100]
n=40
if n in num:
    index=num.index(n)
    print(f"Index of {n} is {index}")
```

```
Index of 40 is 2
```

In [42]:
```
num=[20,30,40,50,70,100]
n=440
if n in num:
    index=num.index(n)
    print(f"Index of {n} is {index}")
else:
    print(f"{n} is not present in list")
```

```
440 is not present in list
```

In [44]:
```
list1=[2,3,'python',100,200,'machine','learning']
n=1000
if n in list1:
    index=list1.index(n)
    print(f"Index of {n} is {index}")
else:
    print(f"{n} is not present in list")
```

```
1000 is not present in list
```

In [45]:
```
list1=[2,3,'python',100,200,'machine','learning']
n='python'
if n in list1:
    index=list1.index(n)
    print(f"Index of {n} is {index}")
else:
    print(f"{n} is not present in list")
```

```
Index of python is 2
```

8)count()

In [ ]:
```
number_of _occurences=List.count(Value)
```

In [48]:
```
list1=[2,3,'python',100,'Python',200,'machine','learning',20,'python','python'
n='python'
list1.count(n)
```

Out[48]: 3

In [49]:
```
list1=[2,3,'python',100,'Python',200,'machine','learning',20,'python','python'
n='Python'
list1.count(n)
```

Out[49]: 1

9)copy()

In [50]:
```
a=50
b=a
print(a)
print(b)
```

```
50
50
```

In [51]:
```
a=50
b=a
print(a)
print(b)

b=b+5
print(a)
print(b)
```

```
50
50
50
55
```

In [52]:
```
x=[4,5,6,7,8]
y=x

print(x)
print(y)
```

```
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]
```

In [53]:
```
x=[4,5,6,7,8]
y=x

print(x)
print(y)

x.append(100)
print(x)
print(y)
```

```
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8, 100]
[4, 5, 6, 7, 8, 100]
```

In [54]:
```
x=[4,5,6,7,8]
y=x  #reference

print(x)
print(y)

y.append(200)
print(x)
print(y)
```

```
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8, 200]
[4, 5, 6, 7, 8, 200]
```

In [55]:
```
x=[4,5,6,7,8]
y=x.copy()
print(x)
print(y)

y.append(200)
print(x)
print(y)
```

```
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8, 200]
```

In [ ]:
```

```